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ANALYSIS OF A LOW-NOISE BOXPROP PROPELLER

Raj Gopala Krishna, Alexander Hansson, Mohammed Kassem, Bala Kumaresh, Oskar Lankamer, Atle Nyberg

TRA370, Uncrewed Aerial Veichles, Group 3
Chalmers University of Technology, Gothenburg, SE-41296, Sweden

ABSTRACT

The work in the report was made at Chalmers as part of the course UAV TRA370.

The boxprop propeller is a propeller design patented by GKN aerospace. The propeller blade consists of two blades next to each other that are joined at the tip. The main advantage of a boxprop is that they create less tip vorticity and can therefore create less noise. Chalmers has previously conducted a lot research on the topic but normally for bigger propellers. The goal of the project was to investigate the aerodynamic and the noise created by a conventional and a boxprop propeller, but on the scale of a small UAV.

The work shows that the boxprop propeller have a slightly worse efficiency than a conventional propeller. Even though the noise of the boxprop couldn't be evaluated in this work it has the capability to reduce the noise levels of a propeller.

NOMENCLATURE

η_p	Propulsive Efficiency
Q	Torque
T	Thrust
n	Revolutions per second
RPM	Revolutions per minute
HTR	Hub to tip ratio
AoA	Angle of attack
CFD	Computational fluid dynamics
SPL	Sound Pressure Level
PSD	Power Spectral Density
BPL	Blade Passage Level
$FW-H$	Ffowcs Williams-Hawkings
DES	Detached Eddy Simulation

1 Introduction

Propellers are often used in small scale UAVs because of their high efficiency at low speed and their simplicity. Propellers can however produce a lot of noise and in some applications that could be problematic. Chalmers has done research on a boxprop propeller with the potential of generating less noise [1]. The purpose of this project is to compare the boxprop blade design to the conventional blade design for smaller scale UAVs. The main question this report will investigate is if the boxprop can lower the noise levels of the propeller without having a significant reduction in efficiency. To make the comparison fair the conventional reference propeller will be a four-bladed propeller and the boxprop a two-bladed propeller as the boxprop is two blades connected at the tips to form one blade. Having a two-bladed normal propeller would mean that it would be loaded more than a two-bladed boxprop and therefore result in an unfair comparison.

2 Approach

To create a reference propeller the zero-dimensional in-house [2] tool written in python named Optoprop was used to generate the geometry. In the next part of the project, BBCode [3] which is a Matlab based tool was adjusted to create the boxprop geometry. Most important output of those two programs were airfoils stacked along the stacking line which were later processed with Python code and imported into Catia V5 using VBA script developed by Dassault Systemes to create the CAD models both for the CFD analysis and manufacturing. Both tools also pro-

vided preliminary aerodynamic analysis of the propellers (zero-dimensional). To assess the actual aerodynamic performance of the propellers, such as efficiency and thrust, steady-state CFD simulations was utilized in STAR-CCM+. Following this, an iterative process was employed to evaluate the propellers. The designs were then refined until the propellers achieved the desired performance. To measure the noise level of the reference propeller CFD simulations in STAR-CCM+ was used in an unsteady flow. Lastly to evaluate the results from both the aerodynamic simulations and noise simulations a wind tunnel test was conducted. In the wind tunnel the efficiency, thrust and noise levels was measured and then later compared to the numerical results.

3 Method

This section presents the methodology of the project and the report. It includes the design process where it's described how the propellers were made and their requirements. It also covers the aerodynamic simulations and how the CFD domain was setup. The noise simulations and how they were conducted is described alongside the methodology behind the wind-tunnel tests.

3.1 Design process

To design the two propellers some requirements needed to be set up. When designing the propellers it was important to keep some of the parameters the same or similar, otherwise the comparison between them could be unfair. One of these requirements was that the hub of the propellers needed to be $45\,mm$ in diameter so they would fit on the motor in the wind tunnel. The motor *RPM* is also limited between 5000 to 12000 *RPM*, therefore a *RPM* of 9500 was picked. Another requirement was that the propellers are gonna travel at $30\,m/s$ and that they generate a thrust of at least $1.5\,N$. A propeller diameter of $0.1524\,m$ was also picked. For the reference propeller an efficiency of at least 75 % needed to be achieved

Hub	45	*mm*
Velocity	30	m/s
Thrust	1.5	*N*
Diameter	0.1524	*m*
Rotational speed	9500	*RPM*
Efficiency	≈ 75	%

TABLE 1. Requirements shared by the two propellers

The design process was similar with both of the propellers. The process started by generating airfoils along a stacking line in either Optoprop or BBcode using the requirement in table 1. The stacking line where then processed in a python code and then imported into Catia V5, with a VBA script, where a 3D CAD model was created. After the 3D model had been created the aerodynamic performance was evaluated with a CFD simulation. Using the calculated thrust and efficiency the propellers could be evaluated and the input parameters in Optoprop or BBcode could be altered. This process was repeated until the blades where significantly loaded and an efficiency of over 75% was achieved.

The reference propeller was created first and therefore set some of the parameters like chord length and thickness. The boxprop was created with this in mind and had similar chord length and thickness. In BBcode there were some additional parameter such as sweep and the chord displacement angle. These were ignored and set to zero as a preliminary design to make the number of variables fewer and as they make most difference for high speed applications. The rest of the parameters, such as AoA and the lift coefficient, were parameters that needed to be iterated upon using CFD to achieve a performance that met the criteria.

3.2 Aerodynamic simulation

The CFD domain was set up by creating two regions, a smaller one from which the propeller geometry was subtracted and a larger domain enclosing it, as can be seen in figure 1. The smaller domain is set in a moving reference frame so that it rotates with the *RPM* of the propeller. For the reference propeller a quarter of the domain was simulated using periodic boundaries on the radial boundaries. While for the boxprop half the domain was needed to simulate it due to it's blade configuration, periodic boundaries were used in the same way. A 3D mesh was used and the k-ω SST model was chosen as the turbulence model which switches between k-ω in the boundary layer and k-ε in the rest if the domain. This is because k-ω is better at resolving the flow in the boundary layer. For the wall treatment the "All y+ Wall treatment" was chosen which uses a low y+ model inside the boundary layer and a high y+ model everywhere else. The equations were solved with the coupled flow setting. To evaluate that the mesh is good enough a mesh convergence study was done where the efficiency was analysed and when the difference was below 1 % the mesh was believed to have converged. With this mesh a mean y+ value on the blade of 0.51 was achieved which is below the targeted 1. A similar procedure was done for the boxprop simulation. The size of the domain was chosen by taking inspiration from the simulation done by Alexandre Capitao Patrao in [1].

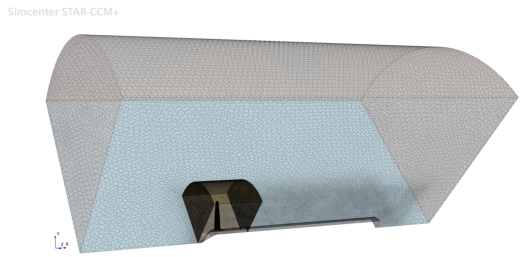


FIGURE 1. Computational Domain for the conventioanl propeller

By measuring the thrust and the torque the propeller efficiency was calculated with:

$$\eta_p = \frac{TV_\infty}{2\pi nQ} \tag{1}$$

3.3 Noise simulation

Aero-acoustic simulations are complex as the sensitive pressure fluctuations are in need to be simulated with appropriate boundary conditions, mesh and solvers settings. This requires higher computational power to capture far-field noise accurately. Initially, experimental data from a study by [4] and a simulation done by [5] of noise generated by unsteady flow over tandem cylinder due to vortex shedding is compared with the simulation results, which is discussed in Section.4.3.1.

The lessons learnt from the former simulations are then applied to analyze the aero-acoustic performance of propellers. The whole domain and all 4 blades was simulated by using mixing plane interfaces for common face between rotating domain and the fluid domain. Unsteady RANS with SST $k-\omega$ turbulence modelling chosen as the model for this simulation. To resolve the boundary layer, the target y+ is set to 1, hence low-y+ wall function is chosen. Coupled flow settings are followed to solve the governing equations. The time step should be small enough to capture the blade passage frequency. The time-step should be 1.75×10^{-5} for 1 degree of rotation of the propeller. Since, due to the time constraint and the computational resources available for this project, a time-step of $4.375 \times 10^{-5} s$ is chosen which means the propeller rotates 2.5 degrees per time-step. The total time period of the simulation is limited to 5 rotations due to computational time and resources. Although the flow variations in the domain becomes periodic after 2 rotations a total of 5 rotations where chosen to ensure convergence. The simulations are verified by evaluating thrust and efficiency with numerical and steady state simulation results.

For Aeroacoustics analysis, FW-H far field noise model is utilized to analyze thickness and loading noise of the propeller. The recever point as shown in Fig.2 is located [0.5, 0.32, 0]m away from the propeller origin. The pressure perturbations is computed for noise levels as a function of time using the FW-H solver. The acoustic time and the noise level is transformed to SPL vs frequency using Point-time Fourier transform function with hamming window function. The amplitude function is set to SPL for the receiver point to receive the noise generated from the blade surface which is considered as the impermeable surface in the FW-H solver settings.

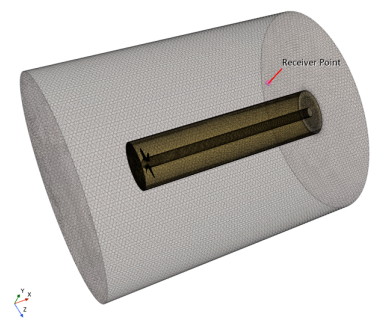


FIGURE 2. Computational Domain for Aero-acoustic Simulation.

As time ran out no Aeroacoustic simulation was done for the boxprop and will therefore be left as a future work. The purpose of comparing two different locations is to ascertain the accuracy of the numerical simulation in capturing noise levels at various locations within the computational domain, particularly at the BPF, a critical indicator of propeller noise. The analysis aims to validate the simulation results due to constraints in computational resources and time, as well as the inability to conduct a comprehensive boxprop aero-acoustics simulation. By relying on the reference propeller results, it becomes imperative to ensure their accuracy before drawing conclusive insights.

3.4 Experimental Investigations

Due to lack of time and that the wind tunnel was not set up the test was not conducted before the deadline. Instead, it took place after the hand in. The purpose of the test it to verify both the performance and the noise levels from the CFD simulations. The propellers were to be 3D printed, attached to an electric motor and spun at the target rpm inside of a low speed wind tunnel. Inside the rig a microphone would be placed to measure the noise levels and the rig would be able to measure both the torque and thrust generated by the propellers. This data would then have been used in the comparison between the two designs.

4 Results

This section presents the results of the project which includes the final design and dimensions of the propellers along with the results from the numerical simulations such as efficiency and noise. Note that there are no experimental results as the wind tunnel test wasn't conducted before the project deadline.

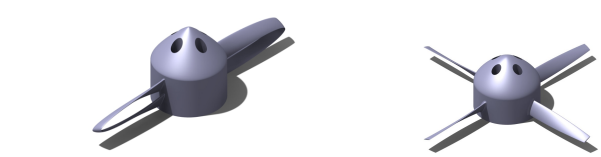
4.1 Propeller designs

The final design of the conventional reference propeller is presented in table 2, and reached an efficiency of 76.22%. Sweeping the propeller over different RPMs gave a peak efficiency of 76.5% at 9000 RPM. This indicates that the propeller is performing close to it's optimal efficiency at it's design advance ratio as seen in figure 2. While the final design of the boxprop achieved a lower efficiency of 70 % and also a lower thrust of 1.6 N compared to the 1.861 N of the conventional propeller. Unlike the conventional one the boxprop achieved it's peak efficiency at 9500 rpm but it's noted that the difference between higher rpm is small for the boxprop as seen in figure 4.

Parameter	Reference	Boxprop	Unit
Advance ratio	1.243	1.243	–
RPM	9500	9500	*RPM*
chord, tip	6.096	8	*mm*
chord, root	10.668	12	*mm*
Thickness, tip	10	12	*% of chord*
Thickness, root	21	20	*% of chord*
Thrust	1.861	1.6	N
CFD efficiency	76.22	70	%

TABLE 2. Reference and boxprop designs

The final CAD designs of the boxprop and conventional propeller is shown in figure 3.



(a) (b)

FIGURE 3. (a) The final CAD design of the Boxprop (b) The final CAD design of the conventional propeller

4.2 Aerodynamic Simulations

The performance of both the reference propeller and the boxprop can be seen in figure 4 where the thrust and power coefficients are plotted along with the efficiency over advance ratio. As can be seen in the figure, the reference propeller outperforms the boxprop in terms of efficiency on all advance ratios except the lowest one where the boxprop has a slightly higher efficiency. Both C_P and C_T are in a similar magnitude especially at the highest and lowest advance ratios. But still there are points in which they differ up to 12 % in C_T which is a notable difference. It can also be noted that for lower RPM the boxprop's efficiency deteriorate faster compared to the reference propeller. While for higher RPM the boxprop's efficiency changes less compared to the reference propeller which rapidly drops. Even though the C_P and C_T is similar at the highest advance ratio and the lowest, the comparison between efficiency differ a lot at these points.

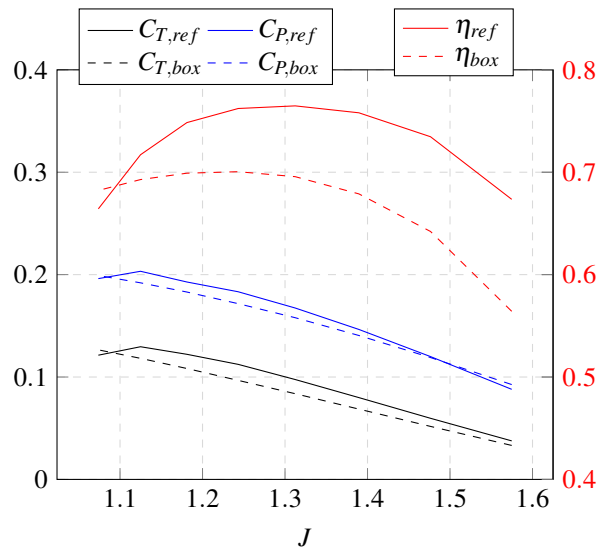


FIGURE 4. The efficiency, C_P and C_T for the boxprop and conventional propeller. Right y axis correspond to efficiency while left correspond to C_P and C_T

Figure 5 shows the pressure distribution from the CFD simulations in STAR-CCM+. They show the pressure around the blade sections at 50 % and 95% of the tip radius. Both propeller designs were evidently producing thrust because of the pressure difference between the suction and pressure sides of the blades. The flow between the blades on the boxprop is accelerated compared to that around the blades, which creates a region of medium pressure compared to a higher pressure at the top of the leading blade and the lower pressure at the trailing blade. This causes an interference between the blades which results in a drop in pressure on the pressure side of the trailing blade. These effects are most notable closer to the tip compared to the root as it's seen in

figure 5 that at 95 % of the tip radius the pressure interference is larger than at 50 % of the tip radius.

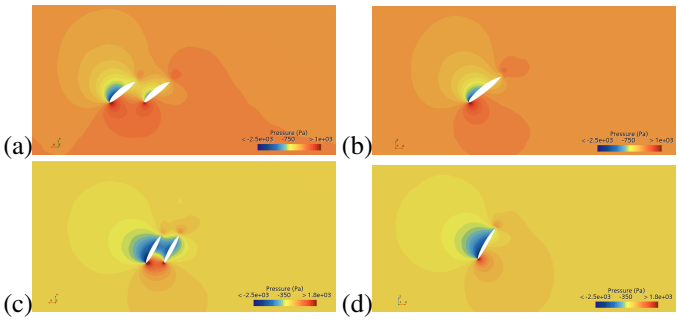


FIGURE 5. (a) Pressure plot of the boxprop at 9500 rpm at 50 % of the tip radius (b) Pressure plot of the conventional propeller at 9500 rpm at 50 % of the tip radius (c) Pressure plot of the boxprop at 9500 rpm at 95 % of the tip radius (d) Pressure plot of the conventional propeller at 9500 rpm at 95 % of the tip radius

In figure 6 the wake of the conventional propeller and the boxprop can be seen. It can be noted that the wake of the conventional propeller is stronger than the one for the boxprop which is a result of a higher thrust. But this would also indicate that the reference propeller would produce more noise from turbulence compared to the boxprop. It's also seen that the wake on the boxprop is stronger closer to the hub while the conventional blade has a more uniform distribution.

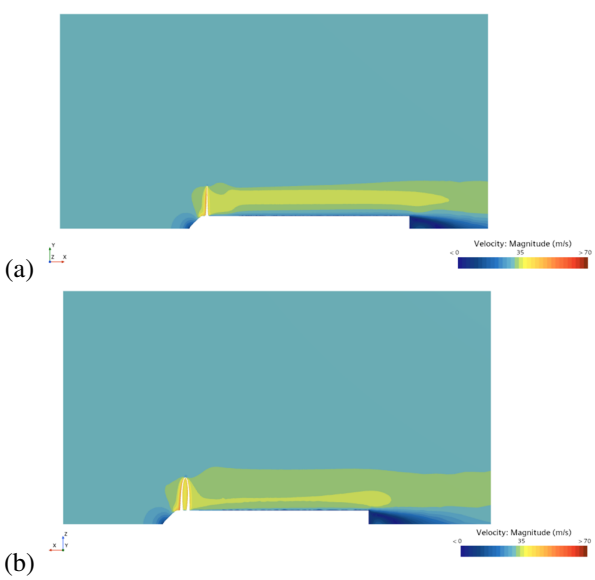


FIGURE 6. (a) The wake of conventional propeller (b) The wake of the boxprop

4.3 Noise simulations

The results for the noise simulations will be presented below.

4.3.1 Tandem Cylinder Figure 7 present aero-acoustic results from experimental data done by [4] and simulations ran by [5], as detailed in the section 3.3. The computational setup follows the study [5]. The results show that the peak magnitude of PSD of Downstream Cylinder from both experimental data (PSD = 140 dB/Hz & Frequency = 181 Hz) and numerical simulation (PSD = 136.926 dB/Hz & Frequency = 165.017 Hz) agrees well with the simulation done in [5] PSD = 133 dB/Hz & Frequency = 183 Hz), but the frequency captured from numerical simulation under-predicts by 0.0884% from experimental results. This deviation is due to insufficient resolution of pressure fluctuations caused by turbulence on the cylinder surface.

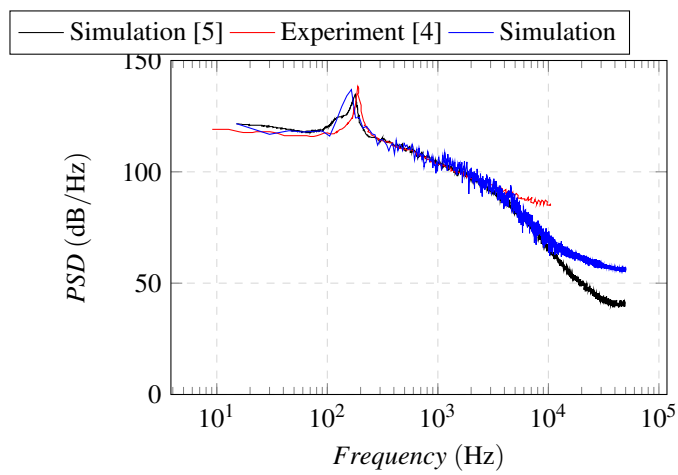


FIGURE 7. PSD of Downstream Cylinder.

4.3.2 Reference Propeller The provided figure, labeled as Figure 8, illustrates the SPL of noise emanating from the propeller, measured both at the receiver point and the blade surface. Observing the figure, it becomes apparent that the highest SPL measured at the blade surface aligns with expectations, occurring precisely at the BPF (123 Hz at 633.33 Hz). However, the discrepancy arises concerning the far-field noise measurements, registering around 57 dB at 1070 Hz.

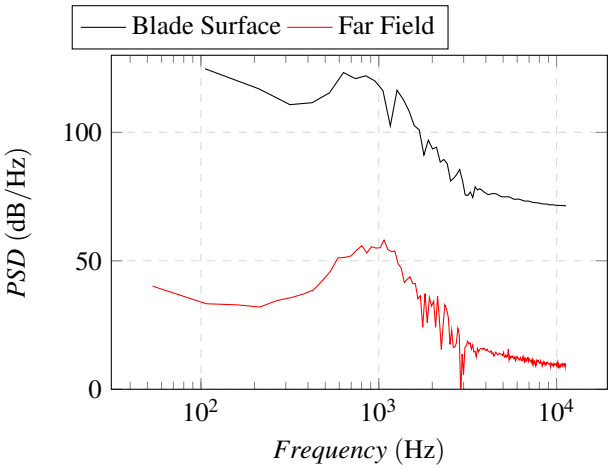


FIGURE 8. The PSD for the Reference Propeller on the blade surface and far field.

5 Discussion and Conclusion

During the project two propellers were designed and produced. The goal was to design both a conventional propeller and compare it to a boxprop to find out if the box-bladed propellers can be used to reduce the noise without a substantial loss in performance. To make the comparison fair the propellers were designed to be as similar as possible to make sure that the physics between the propellers differ minimally. The project resulted in two propellers with similar performance. Due to lack of time noise simulations were only possible for the conventional propeller and wind tunnel tests were delayed beyond the deadline. What has been achieved during this project is the majority of the process to answer the original question but due to lack of time the data needed for the comparison was not attained.

In terms of design the boxprop blades were made thicker than those of the reference due to manufacturing problems while printing the propeller. Due to the increased thickness the blades will be less loaded than the reference. A loaded blade tends to produce more noise which could make a future comparison less fair. For a fair comparison loading of the blade should remain equal between the blades along with chord length as they also differed. Another parameter which could result in a more fair comparison is the length of the blade as one could argue that the tip of the boxprop does not do anything in term of producing thrust as it's orthogonally aligned to the axial direction. Therefore the conventional propeller's blade could be shorter compared to the boxprop to make the effective lift surfaces similar.

The flow accelerating between the box-blades, as shown in figure 5, might cause a loss in thrust. The flow between the blades seem to be accelerating as through a nozzle causing the pressure on the pressure side of the foremost blade to drop and the thrust to decrease. To minimize this loss the blades could have been separated further both axially and radially to decrease the interference between blades. This should especially be made near the tip as it was seen that the most interference occurs at that point. Also in figure 6 it was noted that the wake was weaker near the tip for the boxprop which further points to that it's near the tip that the loss in thrust occurs. In this project the parameters for separation of the blades were set to zero inside the program, any further work would benefit from including them in the design loop.

The different propellers performances were determined through CFD which introduces numerical errors to the comparison, for a more fair comparison both the propellers should have been manufactured and tested in the wind tunnel under the same conditions. The difference between the designs are also apparent both in performance and shape which hinders the process of making a fair comparison. The boxprop had a lower efficiency of about 70% as the reference had an efficiency of around 76% in the operating point, these losses in efficiency could have originated from different flow phenomena such as one propeller separating more at that rotational speed compared to the other and thus different physics could be at play which might make a future comparison of the propellers unfair. It was seen in figure 5 that close to the tip there where a lot of interference between the blades in terms of pressure and that could be the reason for the drop in efficiency. If the span of the project was increased an optimization of both the reference and the boxprop could have been made ensuring a fairer comparison between the two and an increase of efficiency for the boxprop. A wind tunnel test could be conducted to be able to compare both the performance and the noise.

For the noise simulation it could be noted that the tandem cylinders resulted in deviations in the frequency. A solution to be able to capture it better may be achieved by resolving the boundary layer to lower y+ values and using turbulence models such as DES.

From figure 8 it was seen that there was a deviation in the far-field noise measurement. This deviation prompts scrutiny, with potential attributions to the receiver's proximity to the lateral boundary within the computational domain. Such close proximity may induce boundary effects, including reflections or diffractions of sound waves, thereby distorting the measured noise levels. A plausible remedy to mitigate these effects involves the implementation of absorption boundary conditions. These conditions facilitate the absorption of outgoing acoustic

waves, preventing their reflection back into the computational domain and enhancing the accuracy of far-field noise measurements. In conclusion, while the SPL measurements at the propeller blade surface align with expectations, discrepancies in far-field measurements necessitate careful consideration of boundary effects and the implementation of appropriate numerical techniques to ensure accurate simulation results. As no simulation could be done on the boxprop it could not be concluded if the boxprop produces less noise or not. But as it was seen that the tip is producing less thrust compared to the conventional propeller and that because of how the boxblade connects at the tip one would expect that the tip vortices will be less and therefore the noise would be lower for the boxprop.

In conclusion it was shown in this study that there is a possibility of making a boxprop propeller with similar performance to a conventional propeller even though due to lack of time it was not achieved in this study. A CFD simulation to calculate the noise produced for a propeller was achieved and should in future work be used to run the boxprop to calculate the noise of it. But indications where found that a 2 bladed boxprop should produce less noise compared to a 4 bladed conventional propeller. Another future work should be that the propellers be tested in a wind tunnel to assess the real life noise and performance of the propellers.

Contribution

During the project Isak Jonsson and Carlos Xisto was suprevisors for the project and Arion Pons the course coordinator. We would like to thank them for the support and help during this project.
The project also relied on the work of Alexandre Capitao Patrao and Samuel Adriansson. Their work can be found in the reference section.

Alexander Hansson
Responsible for generating the splines in BBCode, also had to do some minor changes with the code. Helped at the start of the project with the CFD of the reference propeller. Report structure and writing.
Atle Nyberg
Have worked on the aerodynamic performance by doing the CFD simulations for both the conventional reference propeller and the boxprop as well as report writing and manufacturing of the 3D printed boxprop.
Mohammed Kassem
Have mainly worked with the aerodynamic performance by doing CFD simulations for both the conventional reference propeller and the boxprop. Has also helped with the design choice of the boxprop along with 3D printing the conventional propeller on FDM printer and report writing.
Oskar Lankamer
Design and optimization of the reference propeller. Both Optoprop and BBCode output handling (python, VBA) and iterating designs for CFD and manufacturing in Catia V5.
Bala Kumaresh
Have conducted a literature survey on noise, performed noise simulations on tandem cylinders, and set up unsteady CFD simulations for the noise analysis of a boxprop. Additionally, wrote comprehensive reports summarizing the findings from these CFD simulations.
Raj Gopala Krishna
Have worked with performing noise simulations on tandem cylinders and unsteady CFD simulations on reference propellers. Report writing on noise analysis.

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